Advanced Materials Characterisation

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Materials Characterisation (A36381)

Short Title:	Adv. Material Characterisation		
Faculty	Science and Computing		
Department	Science		
Cluster	Analytical Science		
Author	KGRENNAN		
Credits	10	**Level:**	Postgraduate

Description of Module / Aims

This module provides an overview of the current technologies used for the characterisation of materials. The principles and instrumentation used in surface analysis, polymorphism, trace analysis and polymer characterisations will be studied. A wide range of surface analysis techniques such as microscopy and micrometric techniques will be covered, specifically their use, sample requirements, principals and limitations. Examples of industrial applications of these techniques will be given, from raw materials and product characterisation to trace analysis and impurity determinations.

Programmes

SCIE-0052	Certificate in Advanced Materials Characterisation (WD_SMATR_MA)	5 / 0 / M
SCIE-0052	MSc in Analytical Science with Quality Management (WD_SANSC_R)	1 / 0 / E
SCIE-0052	PG Dip in Analytical Science with Quality Management (WD_SANSC_G)	1 / 0 / E

Indicative Content

- The properties of surfaces - solids & liquids. Micromeritics to include particle size, specific surface area, porosity, density, surface tension and contact angle measurements.
- Microscopy and imaging in pharmaceutical analysis for determination of morphology and particle size; Optical Microscopy (OM), Scanning Tunnelling Microscopy (STM), Scanning Electron Microscopy (SEM) and Atomic Force Microscopy (AFM)
- Microscopic techniques for contaminant identification, including SEM combined with Energy Dispersive X-ray (EDX) detection
- Particle size analysis techniques based on light scattering
- Thermal analysis and calorimetric techniques, e.g. Differential Scanning Calorimetry (DSC), Thermogravimetric Analysis (TGA) & Dynamic Mechanical Thermal Analysis (DMTA)
- Dynamic Vapour Sorption (DVS) characterisation of pharmaceuticals and moisture determination techniques
- Crystal structure & polymorphism determination techniques, e.g. X-Ray Diffraction (XRD) & thermal microscopy
- Rheology determination techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Establish the principles of operation behind a range of optical, scanning probe and scanning electron microscopic techniques.
2. Establish the principles of operation and applications of a variety of micrometric and thermal analytical techniques for materials characterisation.
3. Establish the principles of a variety of moisture, polymorphism, and particle size and rheology determinations in materials.
4. Evaluate the potential of all the above in the characterisation of materials in the pharmaceutical industrial environment.
5. Plan experiments and calibrate associated laboratory instrumentation for pharmaceutical and biopharmaceutical sample analyses.

Learning and Teaching Methods

- Lectures.
- Practicals.
- Demonstrations.
- E-learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	50%	1,2,3,5
Assignment	50%	1,2,3,4

Assessment Criteria

<40%: The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to practical and resulting data analysis skills.

40%-59%: The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regards to associated practical and resulting data analysis skills.

60%-69%: As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to associated practical and resulting data analysis skills.

70%-100%: All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to associated practical and resulting data analysis skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Practical		24
Independent Learning		222

Essential Material

- "Advanced Materials Characterisation module Moodle page." http://moodle.wit.ie
- "WIT library databases." http://library.wit.ie/Resources/databases

Requested Resources

- Room Type: Chemistry Lab